Training Day 21

1. Topics Covered and Brief Explanation

1. Introduction to Object Detection: Object detection is a computer vision technique used to identify objects in an image or video and determine their locations.
2. Image Classification vs Object Detection:
    - Image classification identifies the main category of an image.
    - Object detection identifies multiple objects and marks their positions within an image.
3. Bounding Boxes: A bounding box is a rectangular box drawn around a detected object to indicate its location.
4. Confidence Score: It represents how confident a model is about its prediction. A higher confidence score indicates stronger confidence in the detected object.
5. Introduction to YOLO: YOLO stands for You Only Look Once. It is a popular real-time object detection algorithm that detects multiple objects in a single image-processing pass.
6. Working of YOLO: YOLO processes an image and predicts object classes, bounding boxes, and confidence scores.
7. Applications of Object Detection: Used in traffic monitoring, surveillance, autonomous vehicles, face detection, and industrial quality inspection.

2. Tools and Technologies Used

Tool/Technology	Purpose
Python	Programming language
OpenCV	Image and video processing
YOLO	Object detection
NumPy	Numerical operations
Google Colab / VS Code	Code execution

3. Practical Learning / Code Examples

A basic object detection example was explored using the Ultralytics YOLO library.

```
# Install the library first:

# pip install ultralytics

from ultralytics import YOLO

# Load a pretrained YOLO model

model = YOLO("yolo11n.pt")
```

```
# Perform object detection
results = model("image.jpg")


# Display detection results
for result in results:
    print(result.boxes)
    result.show()
```

Explanation:

- YOLO("yolo11n.pt") loads a pretrained YOLO11 nano model.
- model() performs object detection on the input image.
- result.boxes provides information about detected objects and their bounding boxes.
- result.show() displays the image with detection results.

4. Learning Outcomes

- Understood the concept of object detection.
- Differentiated between image classification and object detection.
- Learned about bounding boxes and confidence scores.
- Understood the basic working of YOLO.
- Explored practical applications of object detection.